

Chapter 14 (Probability)

Random or non-deterministic experiment:

An experiment whose result cannot be uniquely predicted even if the previous result of the same experiment conducted under similar conditions are all known is called a random or more precisely non-deterministic experiment

A non-deterministic experiment is also sometimes known as a trial.

Sample space and Event:

The totality of all the possible outcomes of an experiment is called the sample space of the experiment. Any component of a sample space is an event.

Equally likely events:

Events are said to be equally likely if there is no valid reason to say that one event has more chance to occur than the others.

Mutually exclusive events:

Events are said to be mutually exclusive if the happening of one forecloses the happening of all the others.

Independent events:

Events are said to be independent if the occurrence of one has no effect on the occurrence of the other or others.

Elementary events:

An elementary event is one which cannot be further subdivided.

Exhaustive set of events:

A set of events is said to be exhaustive if all the possible outcomes are included.

Favourable outcomes:

For every experiment, out of the set of exhaustive outcomes, those entailing the occurrence of a particular event are called the favourable ones for the event.

Classical or Mathematical or a Priori definition of Probability due to Laplace:

Out of n exhaustive, equally likely and mutually exclusive outcomes, if m are favourable to the event A , then the probability of the occurrence of the event A denoted by $P(A)$ is the ratio $m : n$ and we write

$$P(A) = \frac{m}{n} = \frac{\text{Number of outcomes favourable to } A}{\text{Total number of possible outcomes}}$$

Note:

1. $P(A) + P(\bar{A}) = 1$. Here, $\bar{A}$ is the event of non-occurrence of the event A
2. For mutually exclusive events A and B , $P(A \cup B) = P(A) + P(B)$.
3. If A and B are mutually exclusive and exhaustive then $P(A \cup B) = P(A) + P(B)$ and $P(A) + P(B) = 1$
4. For independent events A and B $P(AB) = P(A)P(B)$

Exercise 14.1

1. Give some events in our day-to-day life that are mutually exclusive.

Soln. Events in our day-to-day life that are mutually exclusive are:

- (i) The result of a lottery.
- (ii) The result of throwing a die.
- (iii) The gender of a person.
- (iv) The result of a student's examination.

2. If two events A and B are such that $P(A) + P(B) = 1$, then write $P(B)$ in terms of $P(\bar{A})$.

Soln. We know,

$$\begin{aligned}P(A) + P(B) &= 1 \\ \Rightarrow P(B) &= 1 - P(A)\end{aligned}$$

But,

$$P(A) + P(\bar{A}) = 1.$$

Therefore,

$$P(B) = P(\bar{A}).$$

3. A, B, C are the events which are equally likely but not forming an exhaustive system of events. Then, show that $P(A) = P(B) = P(C) \neq \frac{1}{3}$. Give an example of such a situation.

Soln. Since events A, B, C are equally likely,

$$P(A) = P(B) = P(C) \quad (1).$$

Since events A, B, C are not forming an exhaustive system of events,

$$\begin{aligned}P(A) + P(B) + P(C) &\neq 1. \\ \Rightarrow P(A) + P(A) + P(A) &\neq 1 \quad [\because \text{by (1)}] \\ \Rightarrow 3P(A) &\neq 1 \\ \Rightarrow P(A) &\neq \frac{1}{3}\end{aligned}$$

$$\therefore P(A) = P(B) = P(C) \neq \frac{1}{3}.$$

In throwing a die, if we consider the events of getting 1, 3 and 6 only, they do not form an exhaustive system

But, $P(1) = \frac{1}{6}, \quad P(3) = \frac{1}{6}, \quad P(6) = \frac{1}{6}$

$$\Rightarrow P(1) = P(3) = P(6) = \frac{1}{6}$$

Therefore,

$$P(1) = P(3) = P(6) \neq \frac{1}{3}$$

4. Distinguish between subjective and objective probabilities giving examples in each case.

Soln. Difference between Subjective and Objective Probability

Subjective Probability	Objective Probability
It is based on a person's belief, opinion, or judgement.	It is based on facts, observations, or mathematical calculations.
It may be different for different people.	It is generally the same for everyone when the conditions are the same.
It does not necessarily require a fixed mathematical formula.	It can usually be calculated using a mathematical formula.
Example: A student says, "I think there is a 70% chance that I will win the competition." This is based on the student's personal belief.	Example: When a fair die is thrown, the probability of getting a 6 is $\frac{1}{6}$.

5. When do you say a die is fair?

Soln. A die is said to be fair if all the six faces have equal chances of turning up.

6. When a fair coin is tossed 3 times, how many outcomes can there be? Are the outcomes equally likely?

Soln. Let S be the sample space of throwing a coin three times.

$$S = \{HHH, HHT, HTH, THH, TTT, TTH, THT, HTT\}.$$

Total number of possible outcomes is 8.

Yes, all the outcomes are equally likely.

7. From an urn containing 4 white and 5 red balls two balls are drawn at random. Find the probability that at least one is white.

Soln.

Here, Number of white balls = 4,

Number of red balls = 5,

Total number of balls = $4 + 5 = 9$.

The probability of drawing a red ball in the first draw, $P(A) = \frac{5}{9}$

The probability of drawing a red ball in the second draw (given that the first ball was red),

$$P(B) = \frac{4}{8} = \frac{1}{2}.$$

Since the events A and B are independent of each other, probability that both the balls are red,

$$P(AB) = P(A) \times P(B) = \frac{5}{9} \times \frac{1}{2} = \frac{5}{18}.$$

Therefore, the probability that at least one ball is white

$$= 1 - P(\text{both balls are red}) = 1 - \frac{5}{18} = \frac{13}{18}.$$

8. From a pack of cards, two cards are drawn at random after a thorough shuffle. Find the probability that both are kings.

Soln. No. of kings in the pack of cards = 4

Total no. of cards = 52

Probability that the first card is king, $P(A) = \frac{4}{52} = \frac{1}{13}$

Then, probability that the second card is king, $P(B) = \frac{3}{51}$

Since both the events A and B are independent, $P(AB) = P(A)P(B) = \frac{1}{13} \times \frac{3}{51} = \frac{1}{221}$

$\therefore$ Probability that both cards are king = $\frac{1}{221}$

9. Two fair dice are rolled. Find the probability that the sum of the points is 7. Also find the sum of points which is the most probable by showing all the possible sums of points in a chart or table.

Soln.

No. of possible outcomes = 36

Then, probability that the sum is 7 = $\frac{6}{36} = \frac{1}{6}$

Sum of points	Favourable outcomes	No. of favourable outcomes	Probability
2	(1,1)	1	$\frac{1}{36}$
3	(1,2), (2,1)	2	$\frac{2}{36}$
4	(1,3), (2,2), (3,1)	3	$\frac{3}{36}$
5	(1,4), (2,3), (3,2), (4,1)	4	$\frac{4}{36}$
6	(1,5), (2,4), (3,3), (4,2), (5,1)	5	$\frac{5}{36}$
7	(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)	6	$\frac{6}{36}$
8	(2,6), (3,5), (4,4), (5,3), (6,2)	5	$\frac{5}{36}$
9	(3,6), (4,5), (5,4), (6,3),	4	$\frac{4}{36}$
10	(4,6), (5,5), (6,4),	3	$\frac{3}{36}$
11	(5,6), (6,5)	2	$\frac{2}{36}$
12	(6,6)	1	$\frac{1}{36}$

$\therefore$ The most probable sum of points as observed from the table is 7

10. Given that p is the probability that a person aged x years will die in a year, find the probability that none of the four persons all aged x years will die in a year.

Soln. Let A be the event that a person aged x years will die in a year

Let B be the event that a person aged x years will live in a year

We know that A and B form an exhaustive set of events.

Then, $P(A) + P(B) = 1$

$$\Rightarrow p + P(B) = 1$$

$$\Rightarrow P(B) = 1 - p$$

$\therefore$ The probability that none of the four persons all aged x years will die in a year

$$= (1 - p)(1 - p)(1 - p)(1 - p) = (1 - p)^4$$

11. In the above exercise if Mr. A is one of the four persons, find the probability that at least one of them will die in a year and Mr. A is the first person to die.

Soln. Let A be the event that all of them live and B be the event that at least one of them die in a year.

Then, events A and B form an exhaustive set

$$\Rightarrow P(A) + P(B) = 1$$

$$\Rightarrow (1 - p)^4 + P(B) = 1 \quad [from \text{ question } 10]$$

$$\Rightarrow P(B) = 1 - (1 - p)^4$$

Probability that Mr. A is the first person to die = $\frac{1}{4}$

$\therefore$ Probability that at least one of them will die in a year and Mr. A is the first person to die

$$= \frac{1}{4} \times [1 - (1 - p)^4] = \frac{1 - (1 - p)^4}{4}$$

12. From a well-shuffled pack of cards, two cards are drawn at random. Find the probability that both the cards are diamonds.

Soln.

Number of cards in a pack of cards = 52.

Number of diamonds = 13.

Let A be the event of drawing a diamond in the first draw.

Let B be the event of drawing a diamond in the second draw.

Then,

$$P(A) = \frac{13}{52} = \frac{1}{4}.$$

After the first draw, the number of diamonds left is 12 and the total number of cards left is 51.

Therefore,

$$P(B) = \frac{12}{51} = \frac{4}{17}.$$

Hence, probability that both cards are diamonds,

$$P(AB) = P(A)P(B) = \frac{1}{4} \times \frac{4}{17} = \frac{1}{17}.$$

13. A letter is chosen at random from the word "ASSASSINATION". Find the probability that the letter is (i) a vowel (ii) a consonant.

Soln.

Number of letters in the word ASSASSINATION = 13.

Number of vowels = 6.

Number of consonants = 7.

(i) The probability of choosing a vowel is $\frac{6}{13}$.

(ii) The probability of choosing a consonant is $\frac{7}{13}$.

14. In a 20-20 cricket match, a batsman hits a boundary 5 times out of 24 balls he faced. Find the probability he did not hit a boundary in a ball he faced.

Soln.

Total number of balls faced by the batsman = 24.

Number of boundaries hit by the batsman = 5.

The probability of hitting a boundary in a ball faced by the batsman is

$$\frac{5}{24}.$$

Therefore, the probability of not hitting a boundary is

$$1 - \frac{5}{24} = \frac{19}{24}.$$

15. A dice is thrown. Find the probability of the following events:

- (a) A prime number will appear.
- (b) A number less than 6 will appear.
- (c) A number more than 6 will appear.
- (d) A number less than or equal to 3 will appear.

Soln. Total number of possible outcomes in throwing a die = 6

- (a) Prime numbers are 2, 3, 5.

Number of favourable cases = 3.

Hence,

$$P(\text{prime number}) = \frac{3}{6} = \frac{1}{2}.$$

- (b) Numbers less than 6 are 1, 2, 3, 4, 5.

Number of favourable cases = 5.

Hence,

$$P(\text{number less than 6}) = \frac{5}{6}.$$

- (c) There is no number greater than 6 on a die.

Number of favourable cases = 0.

Hence,

$$P(\text{number greater than 6}) = \frac{0}{6} = 0.$$

- (d) Numbers less than or equal to 3 are 1, 2, 3.

Number of favourable cases = 3.

Hence,

$$P(\text{number} \leq 3) = \frac{3}{6} = \frac{1}{2}.$$

16. There are four men and three ladies in a council. If two council members are selected at random for a committee, how likely is it that both are ladies?

Soln. Let A be the event of choosing a lady in the first selection.

Let B be the event of choosing a lady in the second selection.

Number of ladies = 3.

Number of men = 4.

Total number of persons = $4 + 3 = 7$.

Therefore,

$$P(A) = \frac{3}{7}.$$

After the first selection, the number of ladies left is 2 and the total number of persons left is 6.

Therefore,

$$P(B) = \frac{2}{6} = \frac{1}{3}.$$

Thus,

$$P(AB) = P(A)P(B) = \frac{3}{7} \times \frac{1}{3} = \frac{1}{7}.$$

Hence, the probability that both members are ladies = $\frac{1}{7}$.